

AI VIET NAM

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HCMC AI Competition

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STA

Outline

1

Retrieval Task

2

TransNetV2

3

CLIP

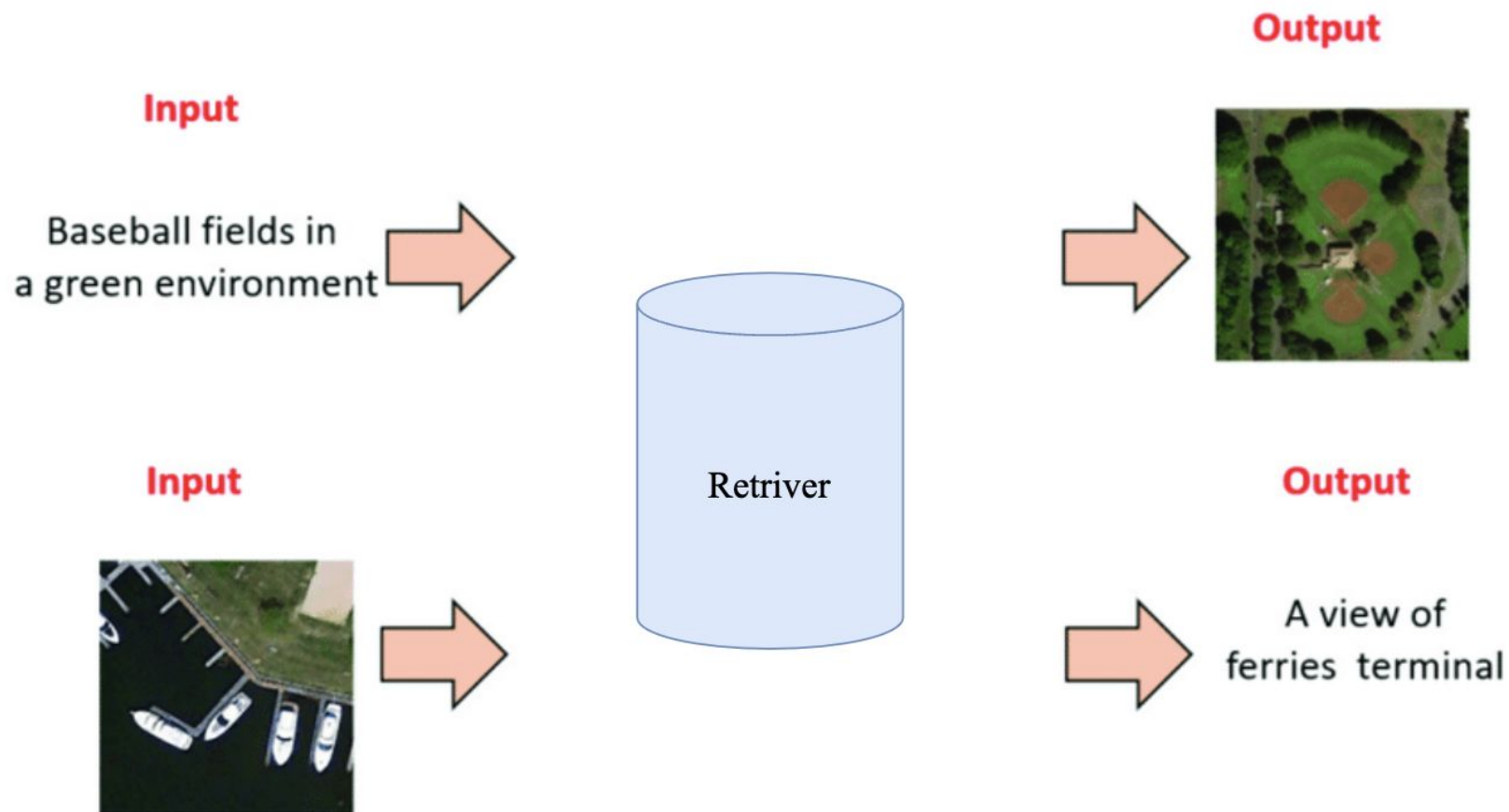
4

FAISS

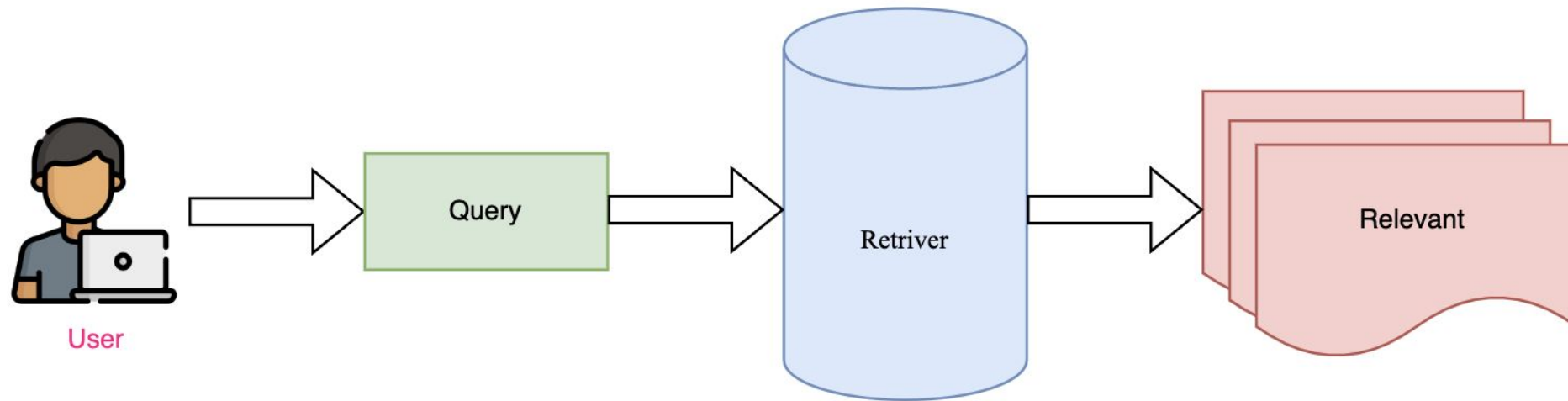
Outline

- 1 Retrieval Task
- 2 TransNetV2
- 3 CLIP
- 4 FAISS

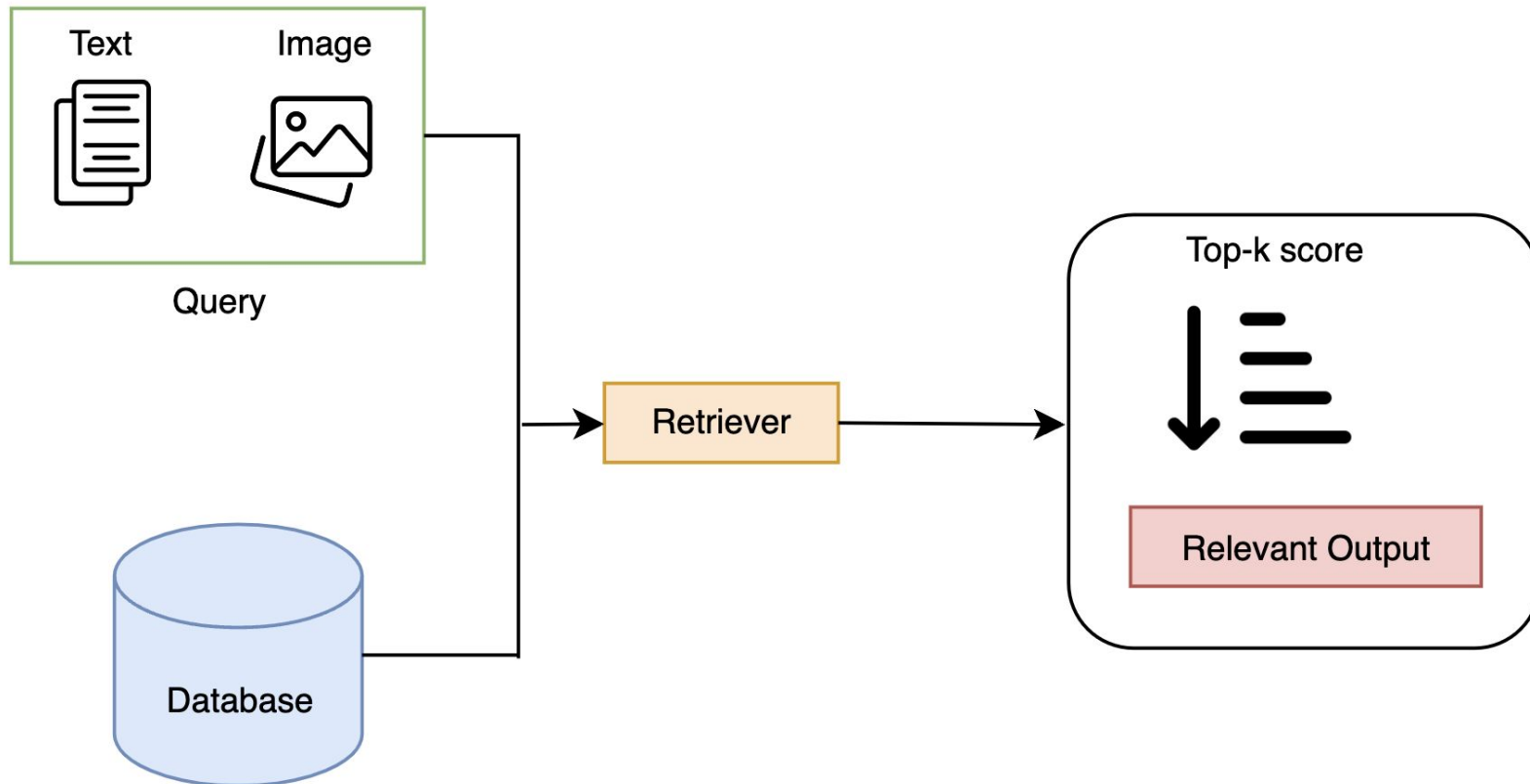
◆ Input-Output



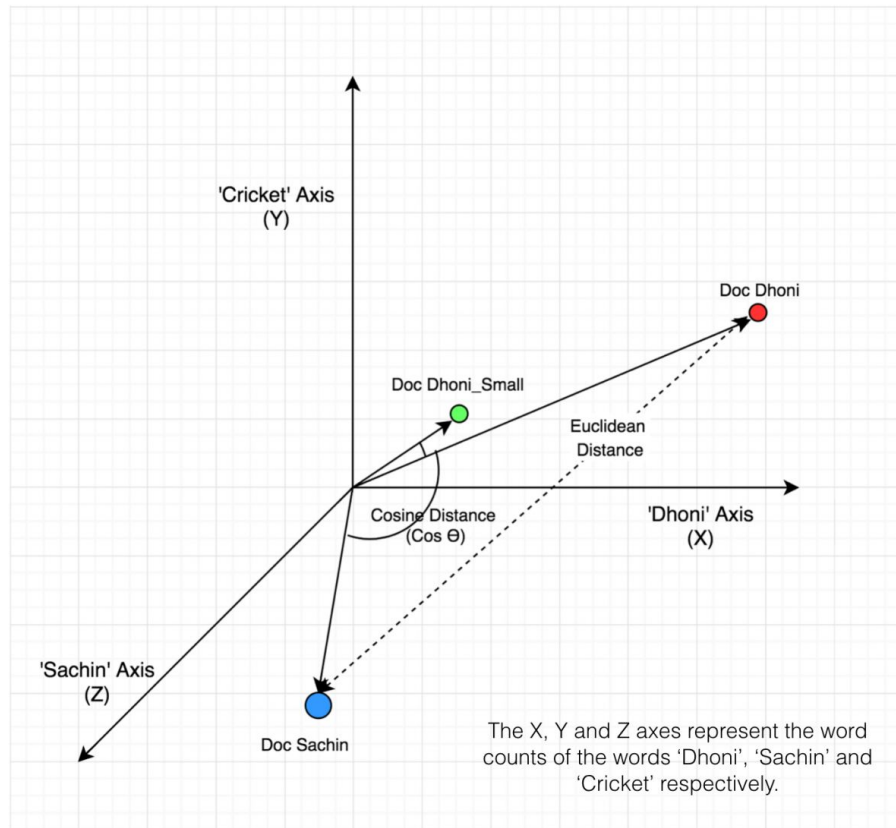
❖ Retrieval



❖ Pipeline



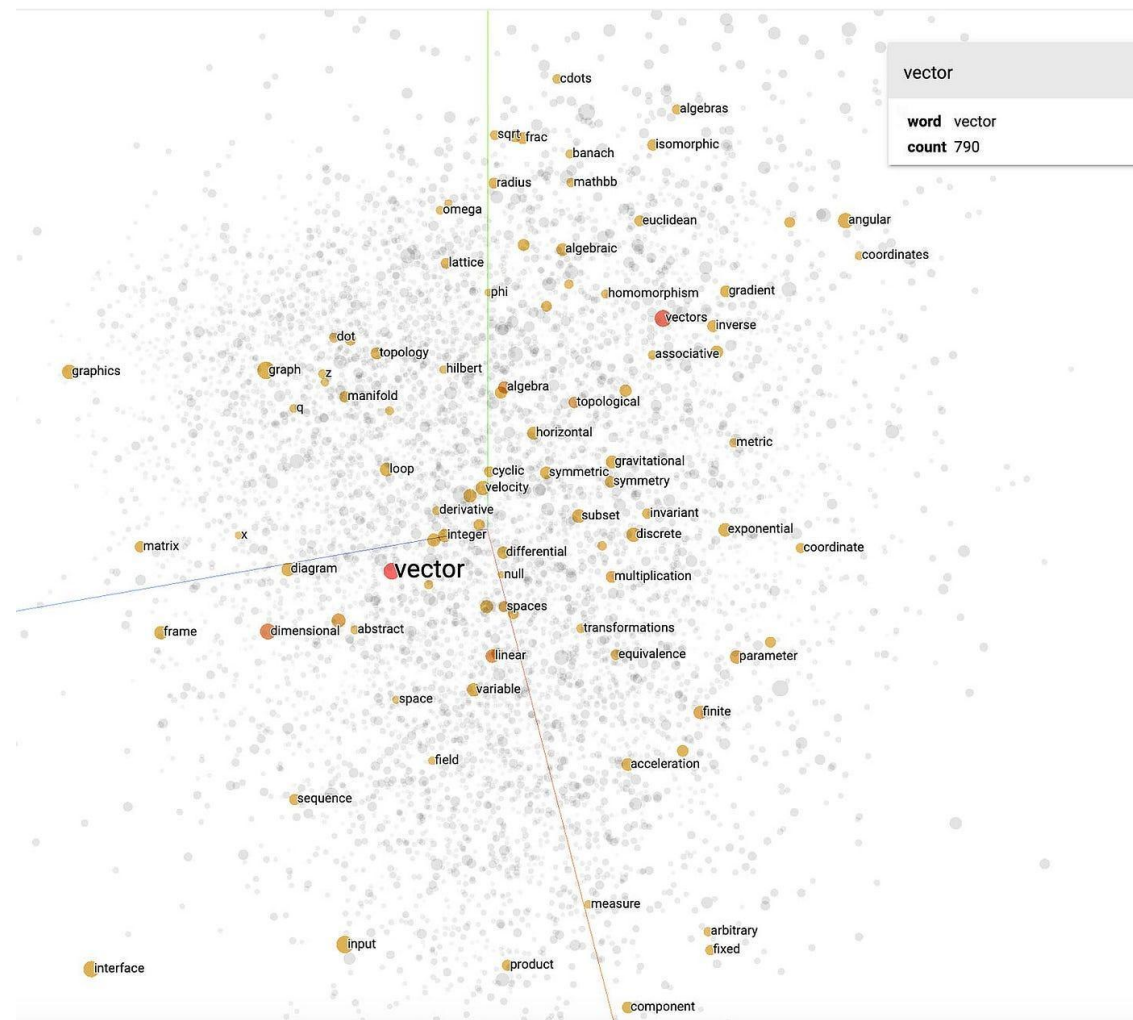
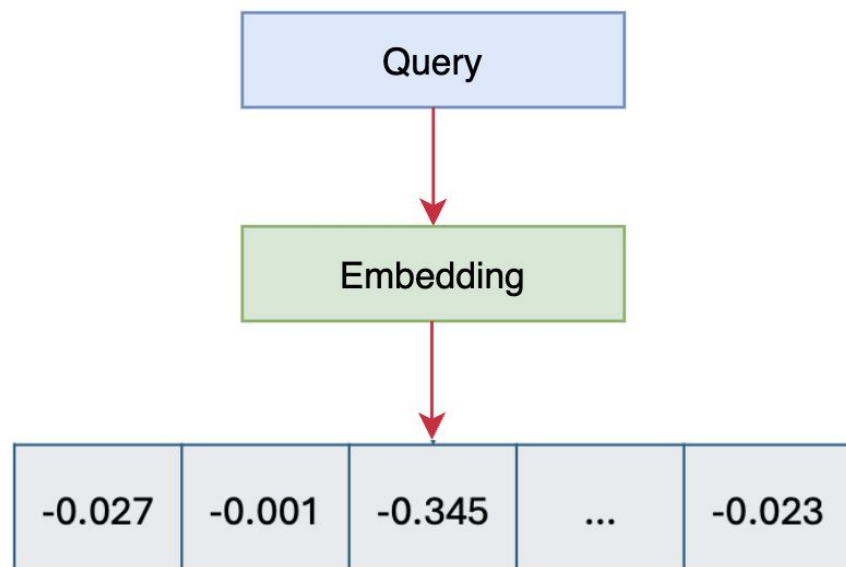
❖ Cosine Similarity



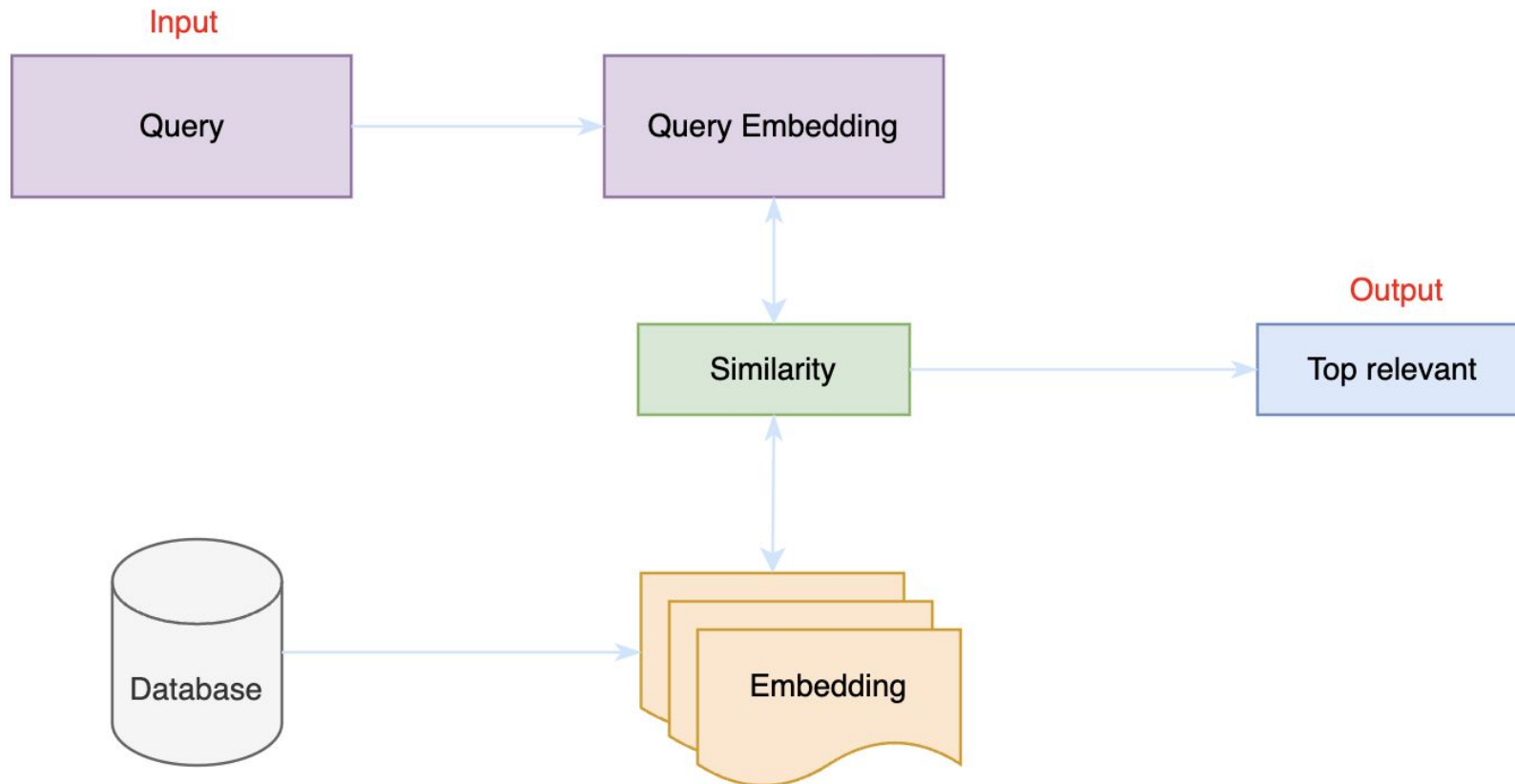
$$\text{similarity}(A, B) = \cos(\theta) = \frac{A \cdot B}{\|A\| \|B\|} = \frac{\sum_{i=1}^n A_i B_i}{\sqrt{\sum_{i=1}^n A_i^2 \sum_{i=1}^n B_i^2}}.$$



Embedding

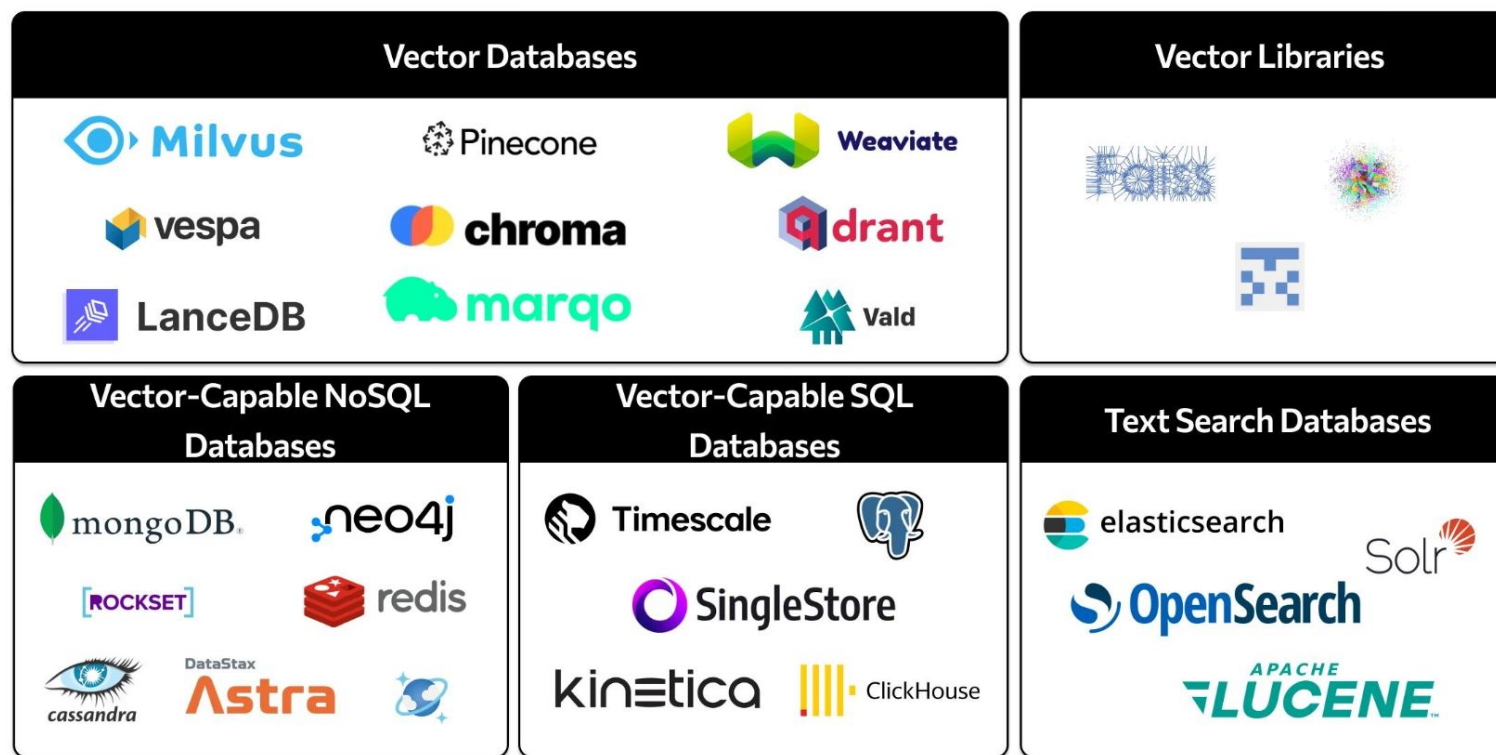


❖ Idea



Vector Database

Choosing a Vector Database



Dataset

- Approximately 1356 videos, each ranging from 10 to 20 minutes in duration.
- Content of the videos focuses on current affairs news updates.
- Includes metadata files and features for each frame.





◆ Dataset/Baseline

Qualifying Round Data

Link: https://docs.google.com/spreadsheets/d/1nnfmplsIB2Cm0jatlN10WjwwZvElvQxctf8_8YKQfXi4/edit?gid=246172816#gid=246172816

Final Round Data

Link :

<https://docs.google.com/spreadsheets/d/16RlhmhETs2YLuW5b4aP-bKD7c0DiDNDr0ZxOQfBS3-ks/edit?gid=1902505481#gid=1902505481>

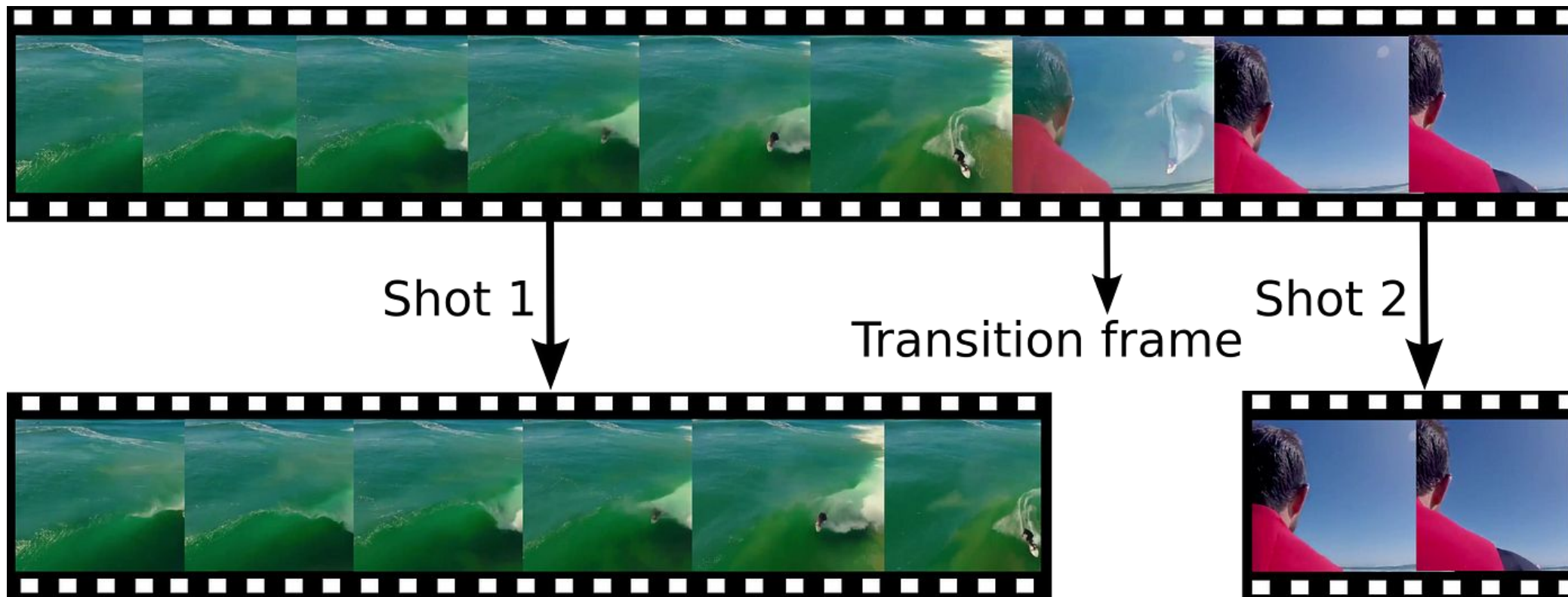
Baseline

GitHub : https://github.com/anminhhung/Pipeline_HCM_AI

Outline

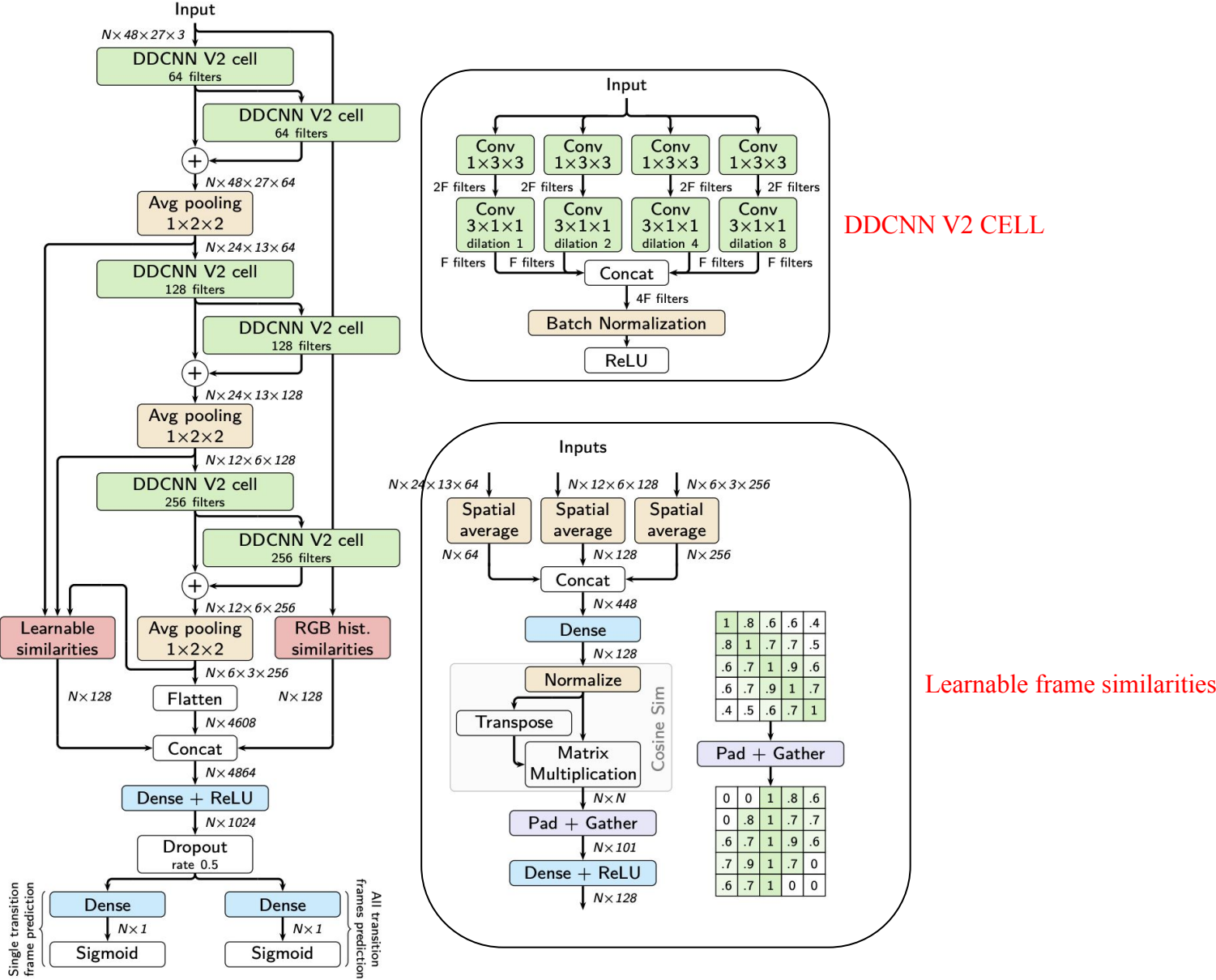
- 1 Retrieval Tasks
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Fast Shot Transition Detection



TransNet V2

Architecture



◆ Prediction



List of scenes (indexed from zero, both limits inclusive):

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Dataset



A computer screen with a Windows message about Microsoft license terms.



A can of green beans is sitting on a counter in a kitchen.



A photo taken from a residential street in front of some homes with a stormy sky above.



A blue sky with fluffy clouds, taken from a car while driving on the highway.



A hand holds up a can of Coors Light in front of an outdoor scene with a dog on a porch.



A digital thermometer resting on a wooden table, showing 38.5 degrees Celsius.



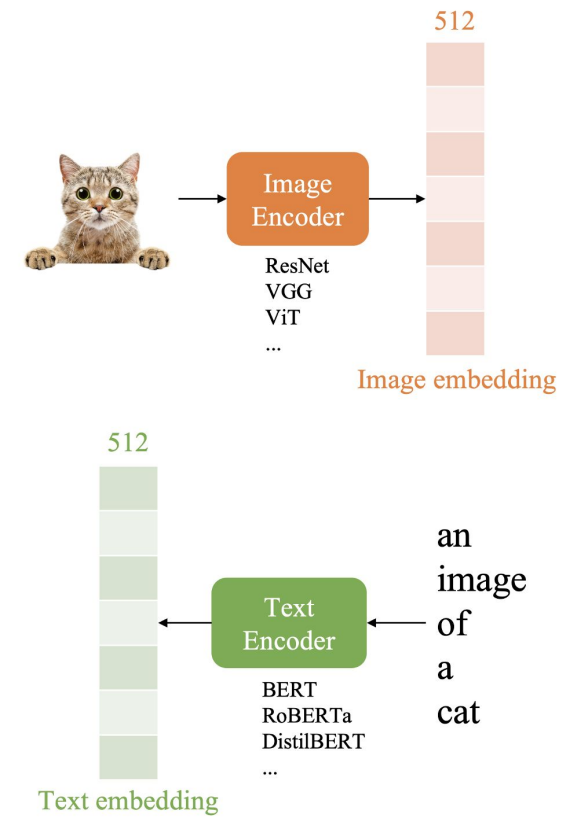
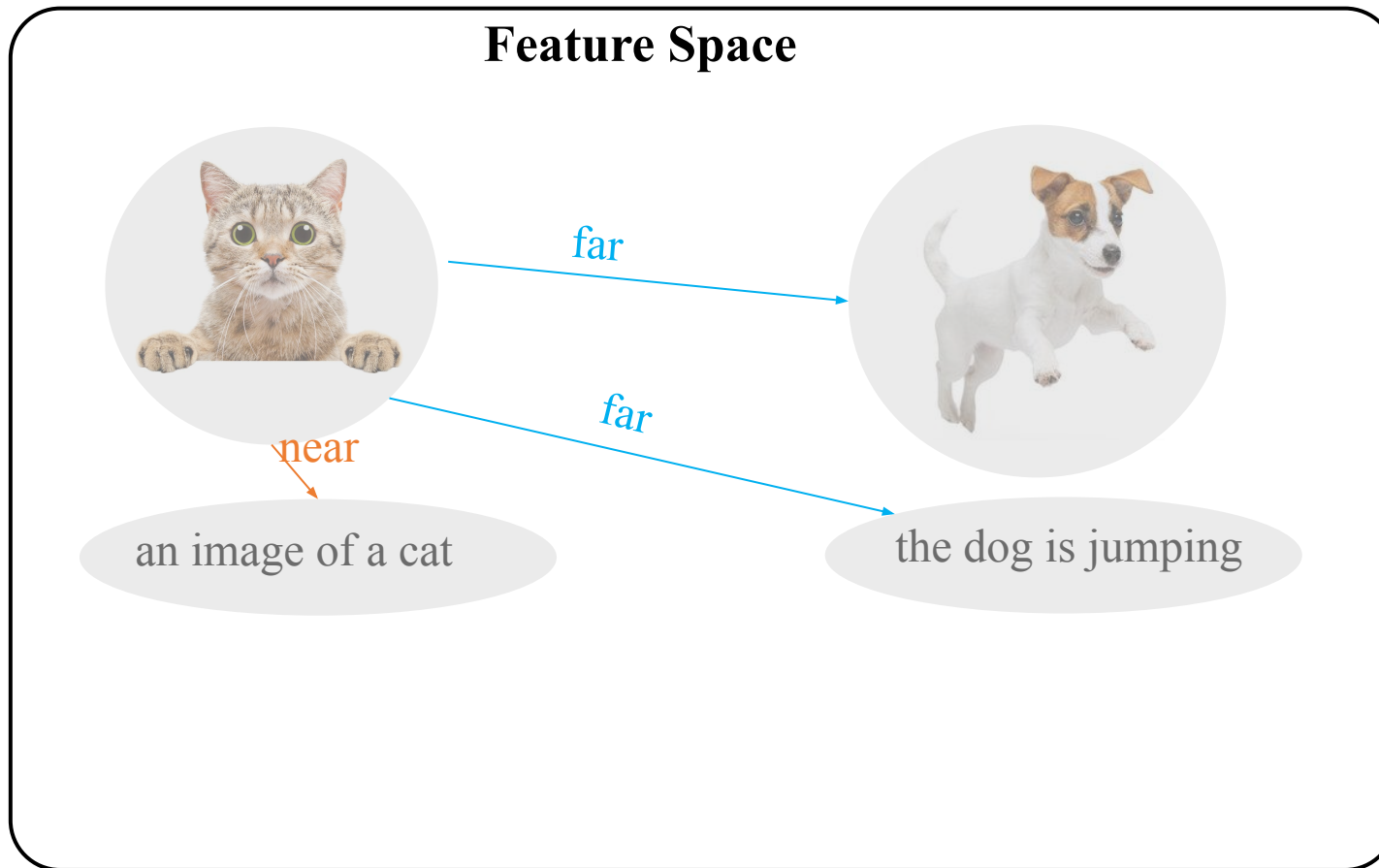
A Winnie The Pooh character high chair with a can of Yoohoo sitting on it in front of a white wall.



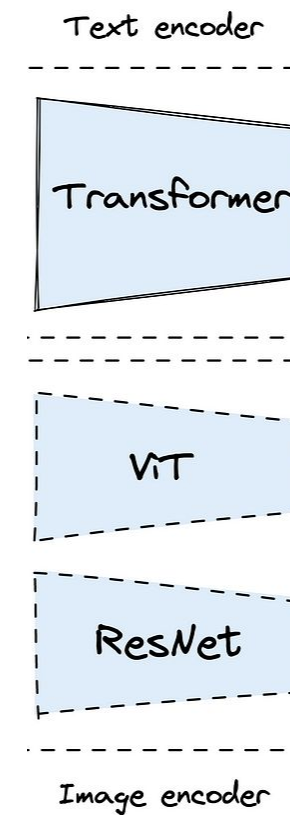
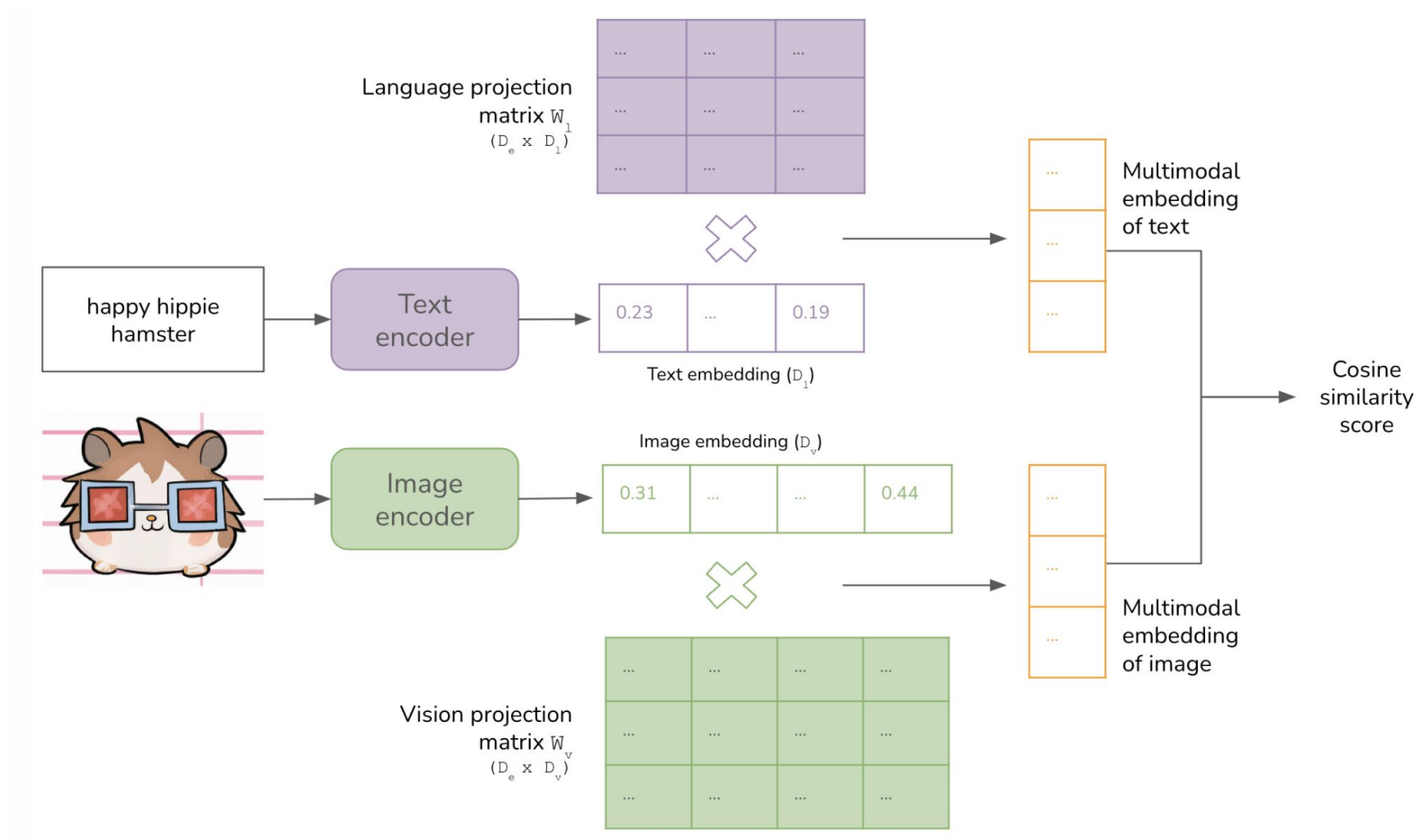
A cup holder in a car holding loose change from Canada.

400M (image, text) pairs collected from Internet

◆ Feature Space

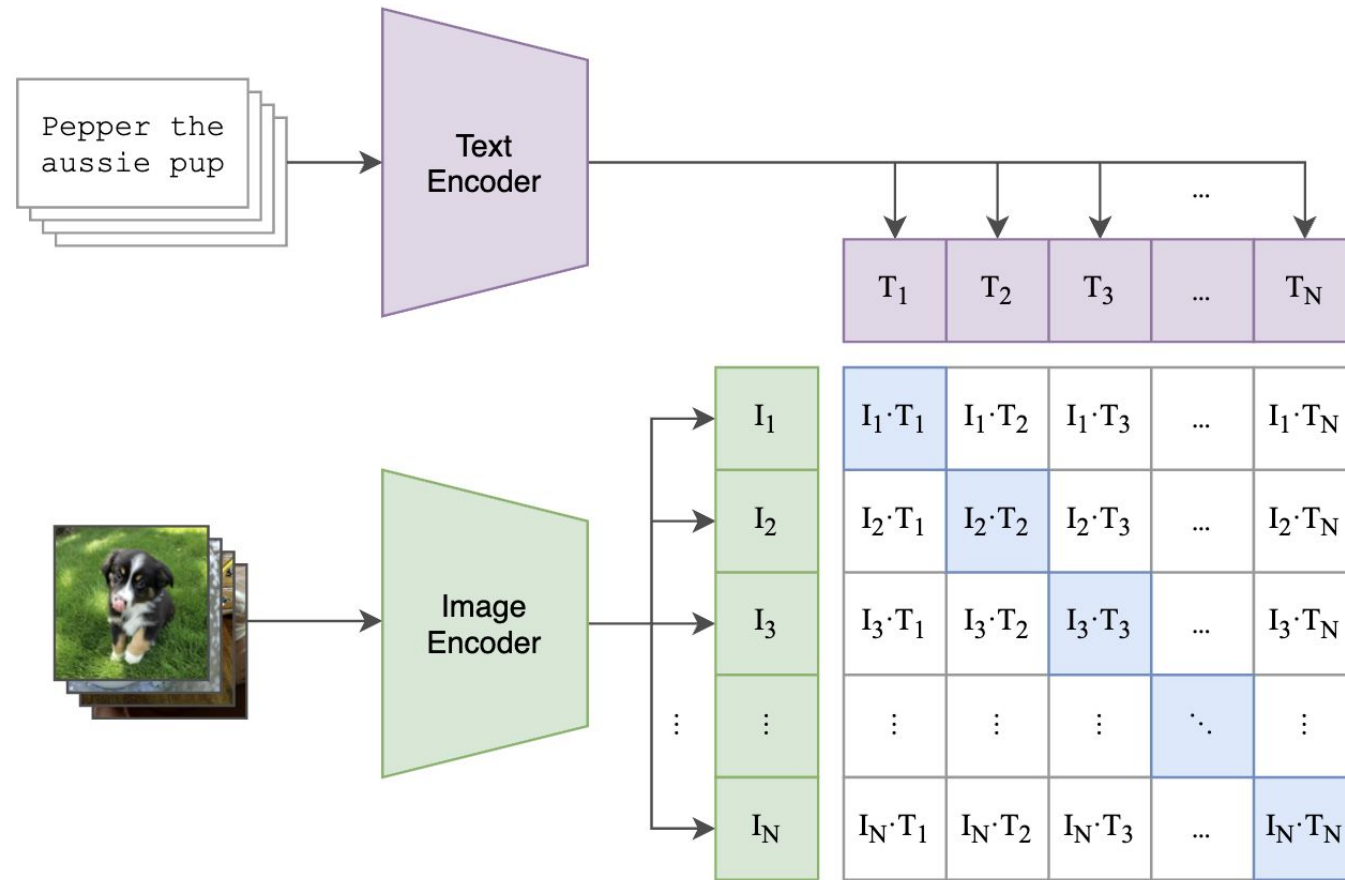


Architecture



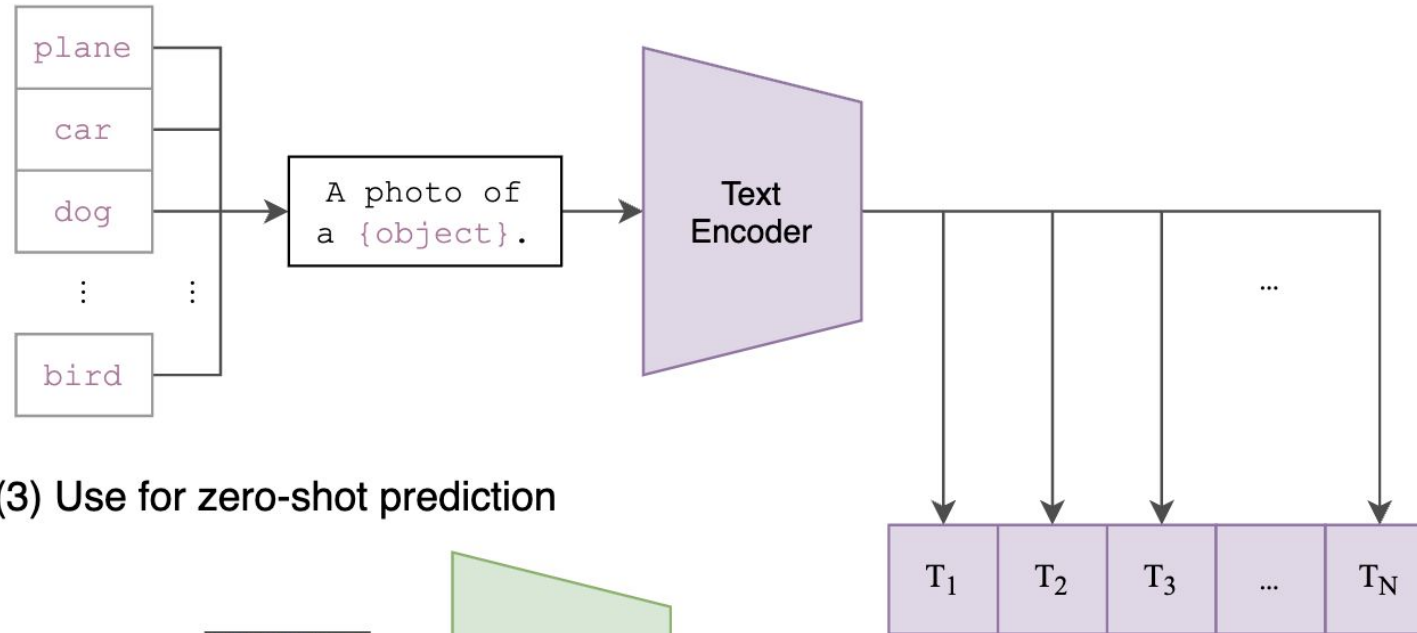
Training

(1) Contrastive pre-training

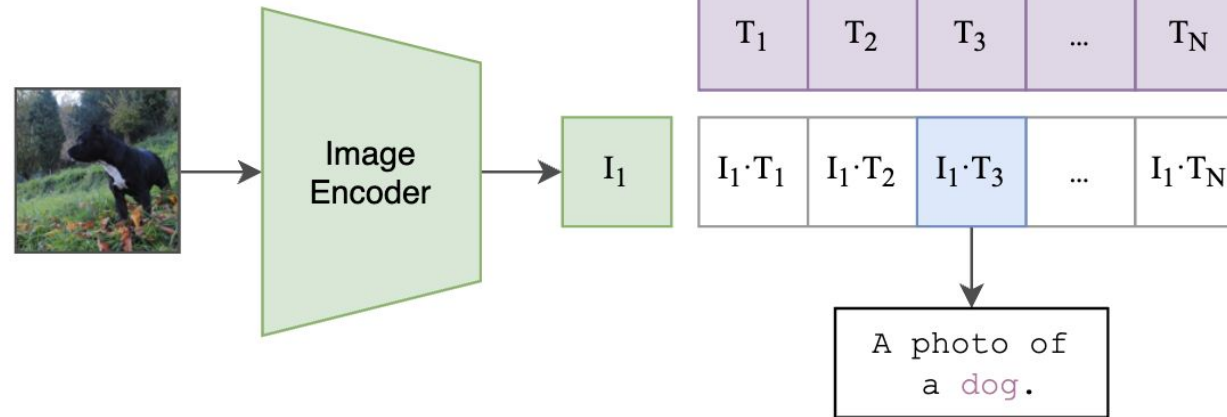


❖ Inference

(2) Create dataset classifier from label text



(3) Use for zero-shot prediction





◆ Inference

Food101

guacamole (90.1%) Ranked 1 out of 101 labels



✓ a photo of **guacamole**, a type of food.

✗ a photo of **ceviche**, a type of food.

✗ a photo of **edamame**, a type of food.

✗ a photo of **tuna tartare**, a type of food.

✗ a photo of **hummus**, a type of food.

Youtube-BB

airplane, person (89.0%) Ranked 1 out of 23 labels



✓ a photo of a **airplane**.

✗ a photo of a **bird**.

✗ a photo of a **bear**.

✗ a photo of a **giraffe**.

✗ a photo of a **car**.

SUN397

television studio (90.2%) Ranked 1 out of 397 labels



✓ a photo of a **television studio**.

✗ a photo of a **podium indoor**.

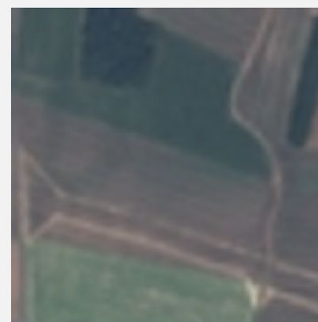
✗ a photo of a **conference room**.

✗ a photo of a **lecture room**.

✗ a photo of a **control room**.

EuroSAT

annual crop land (46.5%) Ranked 4 out of 10 labels



✗ a centered satellite photo of **permanent crop land**.

✗ a centered satellite photo of **pasture land**.

✗ a centered satellite photo of **highway or road**.

✓ a centered satellite photo of **annual crop land**.

✗ a centered satellite photo of **brushland or shrubland**.



Inference

Caltech-101

kangaroo (99.8%) Ranked 1 out of 102 labels



✓ a photo of a **kangaroo**.

✗ a photo of a **gerenuk**.

✗ a photo of a **emu**.

✗ a photo of a **wild cat**.

✗ a photo of a **scorpion**.

Oxford-IIIT Pets

Maine Coon (100.0%) Ranked 1 out of 37 labels



✓ a photo of a **maine coon**, a type of pet.

✗ a photo of a **persian**, a type of pet.

✗ a photo of a **ragdoll**, a type of pet.

✗ a photo of a **birman**, a type of pet.

✗ a photo of a **siamese**, a type of pet.

ImageNet-R (Rendition)

Siberian Husky (76.0%) Ranked 1 out of 200 labels



✓ a photo of a **siberian husky**.

✗ a photo of a **german shepherd dog**.

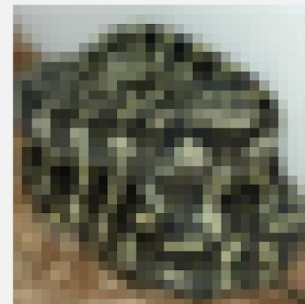
✗ a photo of a **collie**.

✗ a photo of a **border collie**.

✗ a photo of a **rottweiler**.

CIFAR-100

snake (38.0%) Ranked 1 out of 100 labels



✓ a photo of a **snake**.

✗ a photo of a **sweet pepper**.

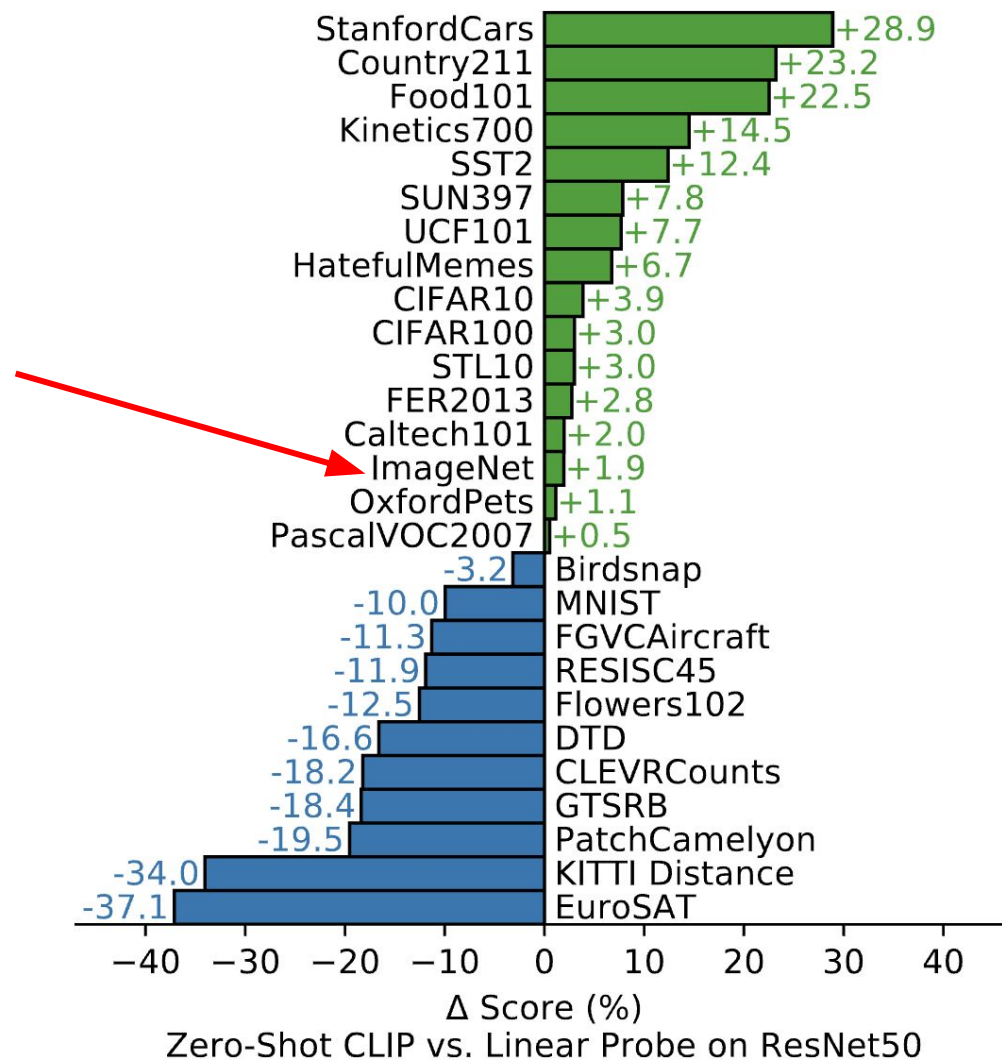
✗ a photo of a **flatfish**.

✗ a photo of a **turtle**.

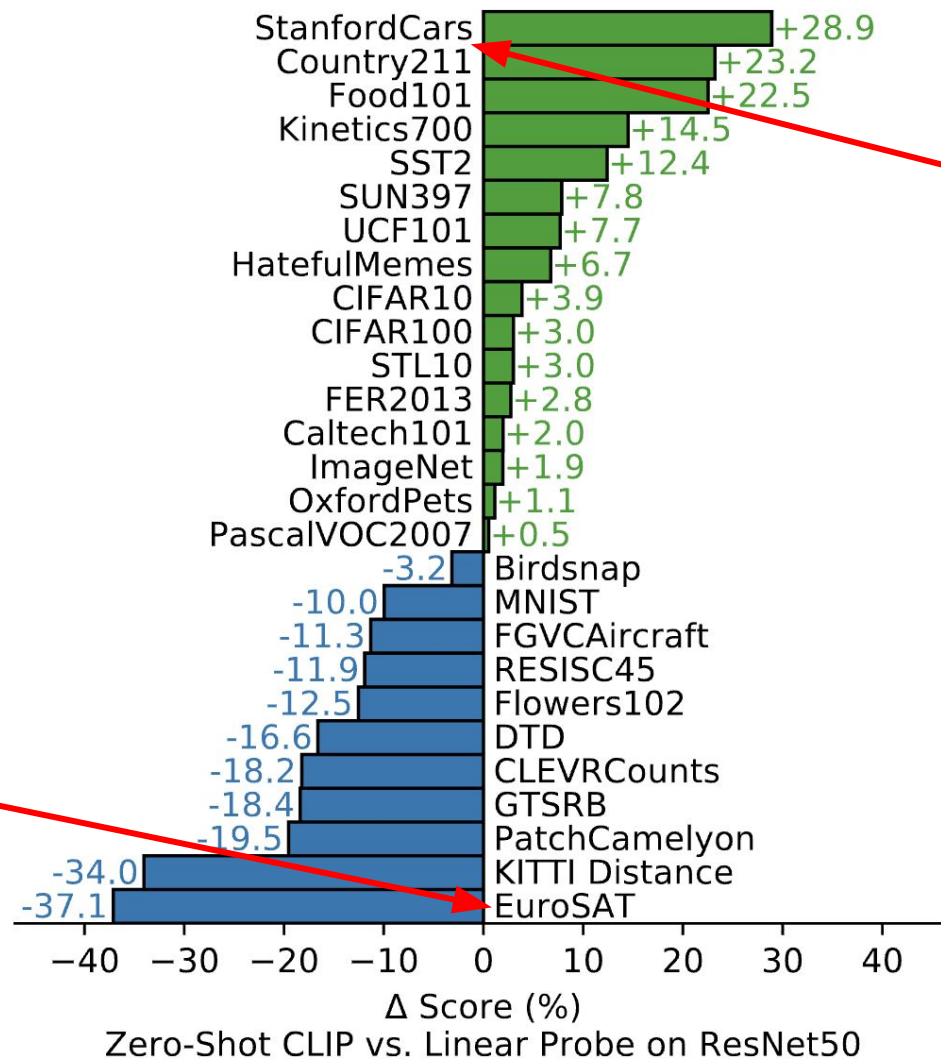
✗ a photo of a **lizard**.



Zero-shot Performance



Zero-shot Performance



❖ Generalize to Out-of-Distribution (OOD)

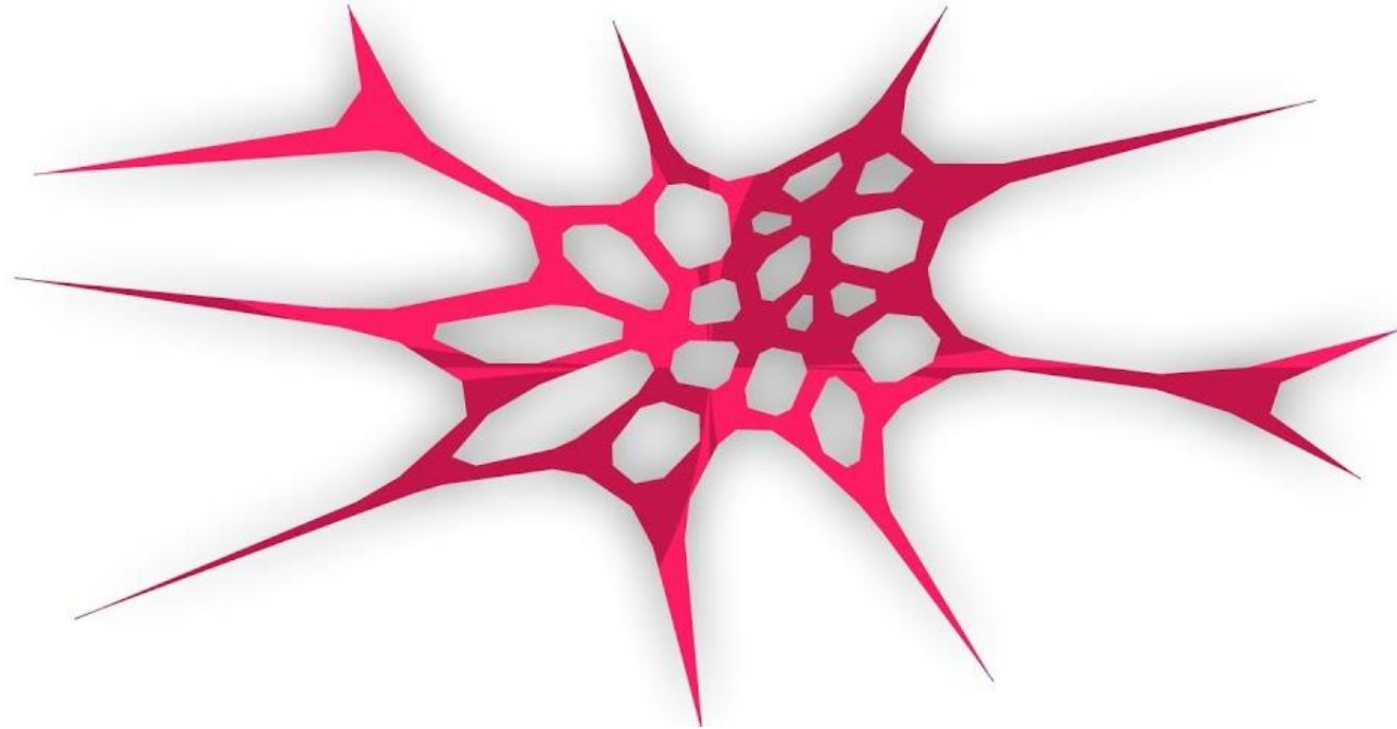
	Dataset Examples	ImageNet ResNet101	Zero-Shot CLIP	Δ Score
ImageNet		76.2	76.2	0%
ImageNetV2		64.3	70.1	+5.8%
ImageNet-R		37.7	88.9	+51.2%
ObjectNet		32.6	72.3	+39.7%
ImageNet Sketch		25.2	60.2	+35.0%
ImageNet-A		2.7	77.1	+74.4%

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FAISS

Scalable Search With Facebook AI



FAISS Pipeline

